Zijun CHEN

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EDUCATION

Hong Kong University of Science and Technology 2022 – Present
PhD in Computer Science (Expected 2027) Hong Kong, China

- **Advisor:** Ke Yi and Nian Si
- **Research Interests:** Reinforcement Learning, Machine Learning Theory, Applied Probability, Game Theory.

Hong Kong University of Science and Technology 2018 – 2022
BSc in Mathematics and Computer Science Hong Kong, China

- **CGA:** 3.938/4.3
- **Honors & Rank:** First Class Honors; Top 5 in the BSc in Mathematics program; Top 1% university-wide at HKUST.
- **Relevant Coursework:** Honors Probability, Advanced Probability Theory, Advanced Mathematical Statistics, Design and Analysis of Algorithms, Numerical Analysis.

PUBLICATIONS AND PREPRINTS

- **Achieving ε^{-2} Dependence for Average-Reward Q-Learning with a New Contraction Principle** 2026
  Zijun Chen, Zaiwei Chen, Nian Si, Shengbo Wang
  arXiv: 2601.21301
- **Sample Complexity of Distributionally Robust Average-Reward Reinforcement Learning** 2025
  Zijun Chen, Shengbo Wang, Nian Si
  Neural Information Processing Systems (NeurIPS) 2025
- **Continual Observation of Joins under Differential Privacy** 2024
  Wei Dong, **Zijun Chen**, Qiyao Luo, Elaine Shi, Ke Yi
  ACM Special Interest Group on Management of Data (SIGMOD) 2024

HONORS, AWARDS, AND FELLOWSHIPS

- Hong Kong PhD Fellowship (HKPFS), *Hong Kong Research Grants Council* 2022 – 2026
- HKUST Academic Achievement Medal (top 1%), *HKUST* 2022
- Chern-Class Achievement Scholarship, *HKUST* 2022
- 15th Epsilon Fund Award, *HKUST, Department of Mathematics* 2020
- University's Scholarship Scheme for Continuing Undergraduate Students, HKUST 2019 – 2022

WORKING EXPERIENCE

WeBank Nov. 2025 – Present
Researcher on Reinforcement Learning Shenzhen, China

- Building an offline RL framework for interest rate adjustment using TD3-BC, BCQ, and CQL on 9M historical decision logs.
- Modeling state/action spaces and reward function, and designing safety constraints aligned with business requirements; augmenting the state with static features and mitigating out-of-distribution challenges.

TEACHING EXPERIENCE

- COMP 3711: Design and Analysis of Algorithms, HKUST — 2025 Spring
- COMP 3711: Design and Analysis of Algorithms, HKUST — 2024 Spring
- COMP 2711: Discrete Mathematical Tools for Computer Science, HKUST — 2023 Spring

PROFESSIONAL SERVICES

Conference Reviewer:

- International Conference on Machine Learning (ICML)
- International Conference on Data Engineering (ICDE)

SKILLS

Computer skills: Python (NumPy/pandas/PyTorch), C++, SQL; Linux, Git; LaTeX
Languages: Chinese (native), English (fluent)

INTERESTS

Sports: Badminton; Half Marathon finisher (1:55); road cycling (Hainan Island loop).